

WMO: 701210

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	

Extreme Annual Design Conditions

Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
Mean				Standard Deviation				n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(<i>a</i>)	(<i>b</i>)	(<i>c</i>)		(<i>e</i>)	(<i>f</i>)	(<i>g</i>)	(<i>h</i>)	(<i>i</i>)	(<i>j</i>)	(<i>k</i>)	(<i>l</i>)	(<i>m</i>)	(<i>n</i>)	(<i>o</i>)	(<i>p</i>)
34.6	31.2	27.6	DB	-47.2	73.0	6.5	4.7	-51.8	76.4	-55.6	79.1	-59.2	81.8	-63.9	85.2
			WB	-32.8	60.9	2.9	3.2	-34.9	63.2	-36.5	65.1	-38.2	66.9	-40.3	69.2

(5)	WB	-32.8	60.9	2.9	3.2	-34.9	63.2	-36.5	65.1	-38.2	66.9	-40.3	69.2	(5)
-----	----	-------	------	-----	-----	-------	------	-------	------	-------	------	-------	------	-----

Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
--------	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	-13.2	-10.3	-10.7	5.1	26.5	40.8	49.0	45.0	35.6	23.6	5.2	-6.1
	DBStd	25.40	14.44	18.05	15.56	12.47	10.33	7.27	6.44	5.58	6.15	8.15	13.91	14.33
	HDD50	12502	1960	1688	1882	1347	728	290	96	175	433	818	1345	1740
	HDD65	17876	2425	2108	2347	1797	1192	726	495	621	882	1283	1795	2205
	CDD50	99	0	0	0	0	0	14	66	19	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	5	0	0	0	0	0	3	2	0	0	0	0	0
CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	12.2	12.4	12.0	11.7	12.5	13.4	12.2	11.8	12.5	11.7	12.0	11.1	13.3
Precipitation	PrecAvg	17.8	1.3	1.4	1.0	0.8	0.7	0.8	1.9	2.8	2.0	2.4	1.5	1.2
	PrecMax	24.8	2.7	4.4	5.0	2.2	1.8	1.5	3.9	6.1	4.4	4.6	4.6	3.1
	PrecMin	12.8	0.3	0.2	0.2	0.1	0.0	0.4	0.5	0.6	0.5	0.9	0.2	0.4
	PrecStd	3.1	0.6	1.0	0.9	0.6	0.5	0.3	0.8	1.3	0.8	0.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	34.0	32.4	36.8	53.8	68.6	72.6	64.4	56.4	43.6	35.4	32.6
		MCWB	27.0	31.7	31.1	34.0	48.3	57.0	59.6	54.7	49.7	40.5	33.0	31.4
	2%	DB	23.5	29.4	28.0	34.4	45.9	60.9	66.4	60.6	51.7	39.3	32.3	27.6
		MCWB	22.4	28.7	26.8	32.3	42.2	53.0	56.8	53.8	47.6	37.7	31.1	26.4
	5%	DB	16.1	25.4	23.0	28.4	41.1	55.5	63.2	56.1	47.8	36.3	29.0	23.5
		MCWB	15.2	24.1	21.7	27.0	38.6	50.2	55.2	51.8	45.2	34.8	28.2	22.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	32.1	31.5	34.2	47.7	58.4	60.4	57.1	50.9	41.1	33.3	31.8
		MCDB	28.3	33.6	32.6	36.4	53.1	67.4	69.1	61.9	55.0	43.5	34.6	33.4
	2%	WB	23.2	28.9	27.3	32.2	42.2	53.9	58.0	54.4	47.3	37.6	31.2	26.8
		MCDB	24.5	29.8	28.4	34.3	45.3	60.0	65.1	58.9	50.5	39.2	32.3	27.7
	5%	WB	16.2	24.9	22.3	27.2	38.7	50.8	55.8	52.3	45.6	34.6	28.2	22.7
		MCDB	17.1	26.2	23.6	28.4	41.0	55.7	62.0	56.2	47.6	35.9	28.9	24.1
Mean Daily Temperature Range	0.4%	WB	8.1	19.6	14.6	22.0	35.9	47.1	53.8	50.2	42.7	32.6	24.2	14.9
		MCDB	9.3	20.6	15.7	23.2	37.5	51.3	58.9	53.3	45.0	33.7	25.7	15.7
	5% DB	MDBR	12.8	13.2	13.7	13.6	9.4	11.4	12.0	9.8	8.3	9.2	12.1	12.4
		MCDBR	20.3	16.6	15.3	15.0	13.5	19.6	18.8	15.6	14.3	10.3	9.5	15.6
		MCWBR	18.9	16.1	14.9	13.6	10.1	13.3	11.4	10.3	10.6	8.9	9.3	15.2
	5% WB	MCDDBR	19.4	16.6	15.4	15.0	13.4	19.1	17.0	14.4	13.3	10.0	9.3	14.9
MCWBR		18.8	16.1	15.1	13.6	10.2	13.3	10.8	9.8	10.2	8.8	9.1	14.7	
Clear-Sky Solar Irradiance	taub	0.163	0.216	0.247	0.311	0.355	0.337	0.346	0.341	0.304	0.233	N/A	N/A	
	taud	1.906	2.037	2.215	2.088	2.121	2.424	2.481	2.499	2.506	2.192	N/A	N/A	
	Ebn at Noon	9	167	248	257	259	272	265	252	231	160	0	0	
	Edh at Noon	1	13	22	36	39	30	28	24	18	11	0	0	
All-Sky Solar Radiation	RadAvg	4	115	632	1336	1704	1821	1477	964	517	155	0	0	
	RadStd	0	12	36	70	152	140	174	101	46	20	0	0	

	Heating	Cooling	Degree-Days
--	---------	---------	-------------

[illegible]

Nomenclature: See separate page